

Partial Progress on Fully Polynomial Approximation for ProbeMax: A Restricted FPTAS for the Non-Adaptive Constant-Support Case

Research note

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Abstract

Segev and Singla gave efficient polynomial-time approximation schemes for several stochastic probing and stopping problems, including adaptive and non-adaptive PROBE_{MAX}, and asked whether fully polynomial approximation schemes can be obtained. This note does *not* solve that question in full. It gives a restricted positive result for the non-adaptive PROBE_{MAX} problem when the union of all positive support values appearing in the instance has fixed constant size. For every fixed support-union size d , we give a dynamic program, trimmed in the space of threshold-survival probabilities, that returns a set of value at least $\text{OPT} / (1 + \varepsilon)$ in time polynomial in the number of variables, the probing budget, the input bit complexity, and $1/\varepsilon$. The dependence on d is exponential, and hence the result is partial progress rather than an FPTAS for arbitrary-support or adaptive instances.

1 Introduction and status of the result

In the non-adaptive PROBE_{MAX} problem, one is given independent nonnegative random variables $X_1, \dots, X_n$ and an integer budget k . A feasible solution is a set $S \subseteq [n]$ with $|S| \leq k$, and its value is

$$V(S) = \mathbb{E} \left[\max_{i \in S} X_i \right].$$

The goal is to maximize $V(S)$.

The motivation for this note is the fully polynomial approximation question arising from the work of Segev and Singla [1], who obtain EPTAS results for stochastic probing and stopping problems including adaptive and non-adaptive PROBE_{MAX}. The result below is deliberately narrower. It addresses only the non-adaptive set-selection problem and only under a constant global support-union assumption.

Status. The theorem proved here is a genuine FPTAS for a special case of non-adaptive PROBE_{MAX}. It is not a solution of the general FPTAS question for Segev–Singla’s problems. In particular, it does not handle arbitrary support size, adaptive probing policies, Free-Order Prophets, Pandora’s Box with Commitment, or other stopping variants.

2 Model

The input consists of independent nonnegative random variables $X_1, \dots, X_n$, each given by an explicit finite rational distribution, and a budget k . We may assume $k \leq n$ by replacing k with $K = \min\{k, n\}$; throughout the proof K denotes this truncated budget. Let

$$0 = v_0 < v_1 < \dots < v_d$$

be the union of all positive support values appearing in the instance. The parameter d is treated as a fixed constant. For each threshold $\ell = 1, \dots, d$, define

$$\Delta_\ell = v_\ell - v_{\ell-1}, \quad p_{i\ell} = \mathbb{P}[X_i \geq v_\ell].$$

For a set $S \subseteq [n]$, define the threshold-survival profile

$$r_\ell(S) = \mathbb{P}\left[\max_{i \in S} X_i \geq v_\ell\right] = 1 - \prod_{i \in S} (1 - p_{i\ell}).$$

The product formula uses independence. We write

$$r(S) = (r_1(S), \dots, r_d(S)) \in [0, 1]^d$$

for the full profile.

3 Main theorem

Theorem 1 (Restricted FPTAS). *For every fixed integer $d \geq 1$, there is an FPTAS for non-adaptive PROBLEM MAX on instances whose union of positive support values has size at most d . More precisely, given $\varepsilon \in (0, 1]$, the algorithm below returns a feasible set S such that*

$$V(S) \geq \frac{\text{OPT}}{1 + \varepsilon}, \quad \text{OPT} = \max_{|T| \leq k} V(T).$$

For fixed d , the running time is polynomial in n , K , the rational input bit complexity, and $1/\varepsilon$. More explicitly, the number of dynamic-programming states is at most

$$(K + 1) \left(1 + \frac{4dnK}{\varepsilon}\right)^d,$$

and the total running time is

$$O\left(nK \left(1 + \frac{4dnK}{\varepsilon}\right)^d \cdot \text{poly}(n, L, \log(1/\varepsilon))\right),$$

where L denotes the bit complexity of the rational input.

The rest of the note proves Theorem 1.

4 Survival-profile representation

Lemma 1 (Discrete tail formula). *For every feasible set S ,*

$$V(S) = \sum_{\ell=1}^d \Delta_\ell r_\ell(S).$$

Proof. Let $Y = \max_{i \in S} X_i$. Since every X_i is supported on $\{0, v_1, \dots, v_d\}$, so is Y . For any nonnegative random variable Y supported on this ordered set,

$$\mathbb{E}[Y] = \sum_{\ell=1}^d (v_\ell - v_{\ell-1}) \mathbb{P}[Y \geq v_\ell].$$

Applying this identity to $Y = \max_{i \in S} X_i$ gives the desired formula, because

$$\mathbb{P}[Y \geq v_\ell] = 1 - \mathbb{P}[X_i < v_\ell \text{ for every } i \in S] = 1 - \prod_{i \in S} (1 - p_{i\ell}) = r_\ell(S).$$

□

It is useful to define the linear value functional

$$\Phi(r) = \sum_{\ell=1}^d \Delta_\ell r_\ell.$$

By Lemma 1, $\Phi(r(S)) = V(S)$.

5 The trimmed dynamic program

If $K = 0$ or every variable has expectation zero, the empty set is optimal. Otherwise set

$$B = \max_{i \in [n]} \mathbb{E}[X_i].$$

Then $B > 0$, and since a best single item is feasible,

$$B \leq \text{OPT}. \tag{1}$$

Let

$$\alpha = \frac{\varepsilon}{1 + \varepsilon}.$$

For each threshold ℓ , let

$$M_\ell = \sum_{j=1}^K p_{(j)\ell},$$

where $p_{(1)\ell} \geq p_{(2)\ell} \geq \dots \geq p_{(n)\ell}$ are the probabilities $p_{1\ell}, \dots, p_{n\ell}$ sorted in nonincreasing order. Define

$$U_\ell = \min\{1, M_\ell\}, \quad \eta_\ell = \frac{\alpha B}{2dn\Delta_\ell}.$$

Since $v_0 < v_1 < \dots < v_d$, every Δ_ℓ is positive.

For coordinate ℓ , set

$$m_\ell = \max\left\{1, \left\lceil \frac{U_\ell}{\eta_\ell} \right\rceil\right\}.$$

For $r_\ell \in [0, U_\ell]$, define its cell index by

$$c_\ell(r_\ell) = \min\left\{\left\lfloor \frac{r_\ell}{\eta_\ell} \right\rfloor, m_\ell - 1\right\}.$$

Thus each coordinate cell has diameter at most η_ℓ . For $r \in \prod_\ell [0, U_\ell]$, define

$$c(r) = (c_1(r_1), \dots, c_d(r_d)).$$

The dynamic program processes items in the order $1, 2, \dots, n$. At time t and cardinality q , it stores at most one exact profile vector in each cell. Each stored vector is accompanied by a predecessor pointer, so it is always the exact profile of some actual subset of $\{1, \dots, t\}$ of size q .

The transition for adding item i to a profile r is

$$T_i(r)_\ell = 1 - (1 - r_\ell)(1 - p_{i\ell}) = r_\ell + (1 - r_\ell)p_{i\ell}. \quad (2)$$

The skip transition leaves r unchanged.

Algorithm.

1. Initialize $\mathcal{D}_{0,0}$ with the zero profile $0 \in \mathbb{R}^d$. All other tables are empty.
2. For $i = 1, 2, \dots, n$, create empty tables $\mathcal{D}_{i,q}$ for $0 \leq q \leq K$.
3. For each stored profile $r \in \mathcal{D}_{i-1,q}$, insert the skip candidate r into $\mathcal{D}_{i,q}$.
4. If $q < K$, compute $r' = T_i(r)$ and insert the add candidate r' into $\mathcal{D}_{i,q+1}$.
5. In every insertion, if the destination table already contains a profile in the same cell, keep only the profile with larger value Φ . Store enough predecessor information to reconstruct a corresponding subset.
6. After all variables have been processed, return the subset corresponding to the maximum- Φ profile over all final tables $\mathcal{D}_{n,q}$, $0 \leq q \leq K$.

Because no coordinate is rounded when a profile is stored, every stored profile remains an exact survival profile of its witness subset. The grid is used only to decide which candidates are discarded.

6 Correctness

Lemma 2 (Feasible range and cell count). *For every feasible set S and every threshold ℓ ,*

$$0 \leq r_\ell(S) \leq U_\ell.$$

Moreover,

$$\frac{U_\ell}{\eta_\ell} \leq \frac{2dnK}{\alpha}.$$

Consequently,

$$m_\ell \leq 1 + \frac{2dnK}{\alpha}.$$

Proof. The lower bound is immediate. For the upper bound, the union bound gives

$$r_\ell(S) = \mathbb{P} \left[\bigcup_{i \in S} \{X_i \geq v_\ell\} \right] \leq \sum_{i \in S} p_{i\ell} \leq M_\ell.$$

Also $r_\ell(S) \leq 1$, so $r_\ell(S) \leq U_\ell$.

For the cell-count estimate, observe that

$$\Delta_\ell p_{i\ell} \leq \sum_{h=1}^d \Delta_h p_{ih} = \mathbb{E}[X_i] \leq B.$$

The equality is Lemma 1 applied to the singleton set $\{i\}$. Summing the K largest values of $p_{i\ell}$ gives

$$\Delta_\ell M_\ell \leq KB.$$

Since $U_\ell \leq M_\ell$,

$$\frac{U_\ell}{\eta_\ell} = U_\ell \frac{2dn\Delta_\ell}{\alpha B} \leq \frac{2dnK}{\alpha}.$$

Finally, $m_\ell = \max\{1, \lceil U_\ell/\eta_\ell \rceil\} \leq 1 + U_\ell/\eta_\ell$. □

Lemma 3 (Non-expansion under item insertion). *For each item i and threshold ℓ , the one-dimensional map*

$$f_{i\ell}(x) = x + (1 - x)p_{i\ell}$$

is 1-Lipschitz on $[0, 1]$.

Proof. For $x, y \in [0, 1]$,

$$|f_{i\ell}(x) - f_{i\ell}(y)| = |(1 - p_{i\ell})(x - y)| \leq |x - y|,$$

since $p_{i\ell} \in [0, 1]$. □

Lemma 4 (Trimming invariant). *After processing the first t variables, for every set $S \subseteq \{1, \dots, t\}$ with $|S| = q \leq K$, the table $\mathcal{D}_{t,q}$ contains a representative profile $\hat{r}$ such that*

$$|\hat{r}_\ell - r_\ell(S)| \leq t\eta_\ell \quad \text{for every } \ell = 1, \dots, d.$$

Proof. The claim is true for $t = 0$, since the only relevant set is the empty set and the zero profile is stored.

Assume the claim holds after processing $t - 1$ variables. Let $S \subseteq \{1, \dots, t\}$ with $|S| = q \leq K$.

First suppose $t \notin S$. By induction, $\mathcal{D}_{t-1,q}$ contains a profile $\tilde{r}$ satisfying

$$|\tilde{r}_\ell - r_\ell(S)| \leq (t - 1)\eta_\ell \quad \text{for every } \ell.$$

The skip transition inserts $\tilde{r}$ as a candidate for $\mathcal{D}_{t,q}$. If it is retained, the desired bound follows immediately. If it is discarded, it is replaced by a profile in the same cell and cardinality class. Since every coordinate cell has diameter at most η_ℓ , the replacement differs from $\tilde{r}$ by at most η_ℓ in coordinate ℓ . Thus the final representative differs from $r(S)$ by at most $t\eta_\ell$ in every coordinate.

Now suppose $t \in S$. Let $S' = S \setminus \{t\}$. By induction, $\mathcal{D}_{t-1,q-1}$ contains a profile $\tilde{r}$ satisfying

$$|\tilde{r}_\ell - r_\ell(S')| \leq (t - 1)\eta_\ell \quad \text{for every } \ell.$$

The add transition inserts $T_t(\tilde{r})$ as a candidate for $\mathcal{D}_{t,q}$. By Lemma 3,

$$|T_t(\tilde{r})_\ell - T_t(r(S'))_\ell| \leq |\tilde{r}_\ell - r_\ell(S')| \leq (t - 1)\eta_\ell.$$

But $T_t(r(S')) = r(S)$ by the exact product formula for survival probabilities. If the candidate is retained, the desired bound follows. If it is discarded, the same-cell replacement can add at most η_ℓ more error in coordinate ℓ . Hence the final representative differs from $r(S)$ by at most $t\eta_\ell$ in every coordinate. □

Lemma 5 (Additive approximation). *The set returned by the dynamic program has value at least*

$$\text{OPT} - \frac{\alpha B}{2}.$$

Proof. Let S^* be an optimal feasible set. By Lemma 4 with $t = n$, the final table contains a representative profile $\hat{r}$ such that

$$|\hat{r}_\ell - r_\ell(S^*)| \leq n\eta_\ell \quad \text{for every } \ell.$$

Therefore

$$\Phi(\hat{r}) \geq \Phi(r(S^*)) - \sum_{\ell=1}^d \Delta_\ell n\eta_\ell = \text{OPT} - n \sum_{\ell=1}^d \Delta_\ell \frac{\alpha B}{2dn\Delta_\ell} = \text{OPT} - \frac{\alpha B}{2}.$$

The algorithm returns a final stored profile of maximum Φ . Each stored profile is the exact profile of its witness subset, so the returned subset has value equal to the returned profile's Φ -value. The claimed additive guarantee follows. $\square$

Proof of Theorem 1. By Lemma 5 and (1), the returned set S satisfies

$$V(S) \geq \text{OPT} - \frac{\alpha B}{2} \geq \left(1 - \frac{\alpha}{2}\right) \text{OPT} \geq (1 - \alpha) \text{OPT} = \frac{\text{OPT}}{1 + \varepsilon}.$$

The last equality uses $\alpha = \varepsilon/(1 + \varepsilon)$.

It remains to bound the running time. By Lemma 2, each coordinate has at most

$$1 + \frac{2dnK}{\alpha} \leq 1 + \frac{4dnK}{\varepsilon}$$

possible cells, since $\varepsilon \in (0, 1]$ implies $\alpha = \varepsilon/(1 + \varepsilon) \geq \varepsilon/2$. Thus the number of profiles stored for each cardinality is at most

$$\prod_{\ell=1}^d m_\ell \leq \left(1 + \frac{4dnK}{\varepsilon}\right)^d.$$

There are $K + 1$ cardinality classes and n item stages, and each stored state generates at most two candidate transitions. This gives the displayed state and arithmetic-operation bounds.

Finally, exact rational arithmetic has polynomial bit complexity. Every stored coordinate is of the form

$$1 - \prod_{i \in W} (1 - p_{i\ell})$$

for some witness subset W of size at most $K \leq n$. Since the input probabilities and support values have rational bit complexity L , all stored coordinates, all values $\Phi(r)$, and all cell-index computations have bit length polynomial in n , L , and $\log(1/\varepsilon)$. Therefore the displayed operation count translates into the stated bit-complexity bound. For fixed d , this is polynomial in n , K , L , and $1/\varepsilon$. $\square$

7 Scope and limitations

The result is meaningful only in the fixed-support-union regime. If the number d of distinct positive support values is part of the input, the factor

$$\left(1 + \frac{4dnK}{\varepsilon}\right)^d$$

is not polynomial in the ordinary input size. Thus Theorem 1 is not an FPTAS for arbitrary-support non-adaptive PROBEMAX.

The proof also uses the fact that a non-adaptive solution is just a set S . It does not give an approximation scheme for adaptive PROBEMAX, where a solution is a decision tree or policy depending on observed outcomes. Nor does it address the stopping/permutation problems studied by Segev and Singla, such as Free-Order Prophets or Pandora’s Box with Commitment.

Corollary 1. *For $\{0, 1\}$ -valued PROBEMAX, and more generally for any class of instances in which all positive support values belong to a fixed finite set of constant size, non-adaptive PROBEMAX admits an FPTAS.*

References

- [1] D. Segev and S. Singla. Efficient approximation schemes for stochastic probing and prophet problems. *arXiv:2007.13121*, 2020. Available at <https://arxiv.org/abs/2007.13121>.